



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 22

Project Title: Linux File Backup and Synchronization System Using System Calls

Team Members

Roll Number	Student Name	Signature
2520090156	Sai Krishna	
2520090218	Navaneeth	
2520090099	Chaithanya	

1. Abstract

The Linux File Backup and Synchronization System Using System Calls is designed to provide a simple and reliable way to back up files and keep two directories synchronized in a Linux environment. Manual file backup can be time-consuming and may result in missing or outdated files. This project addresses the problem by developing a C-based application that performs file operations directly through Linux/POSIX system calls. The system can create a backup of files from a source directory to a backup directory and synchronize changes between directories. Important Operating Systems concepts such as system calls, user space and kernel space, file descriptors, process management, directory handling, file metadata, and permissions are applied. System calls such as `open()`, `read()`, `write()`, `close()`, `stat()`, `mkdir()`, and directory-related POSIX APIs will be used. The proposed solution compares file information and copies updated files to the backup location. The expected outcome is a functional Linux application that demonstrates how system calls can be used for practical file management, backup, and synchronization.

2. Problem Statement

Users often keep important files in a single location, making them vulnerable to accidental deletion, corruption, or loss. Manual copying of files to a backup location is repetitive and may produce outdated or incomplete backups. The project aims to develop a Linux-based file backup and synchronization system using C and system calls. The system will copy files from a source directory to a backup directory and synchronize changed files so that the backup remains updated. This problem is directly relevant to Operating Systems and Systems Programming because it demonstrates file

descriptors, system calls, directory operations, file metadata, permissions, and the transition between user space and kernel space.

3. Objectives

1. Develop a Linux-based file backup application using C system calls.
2. Copy files safely from a source directory to a backup directory using file descriptors.
3. Detect changed files using file metadata and synchronize the backup directory.
4. Demonstrate practical use of Linux system calls, permissions, and user/kernel space interaction.

4. Proposed Methodology

The project will be implemented in C on Linux/Ubuntu. The user will provide a source directory and a backup/synchronization directory through a command-line interface. The program will validate the directories, open source files, read their contents, and write them to corresponding files in the backup directory using Linux system calls. File metadata such as size and modification time will be obtained to identify files that have changed. During synchronization, new or modified files will be copied to the backup location, while existing unchanged files will be skipped. Directory traversal will be handled using POSIX directory APIs. Error handling will be added for missing files, permission problems, and invalid paths. The overall flow is: input paths → scan directories → compare file information → copy changed/new files → update backup → display status and errors.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
open()	Opens source and destination files and returns file descriptors.
read()	Reads file data from the source file into a buffer.
write()	Writes buffered data into the backup file.
close()	Closes file descriptors after file operations are completed.
stat()	Obtains file metadata such as size and modification time for comparison.
opendir()/readdir()	Traverses directories and obtains file names for backup/synchronization.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090156	Sai Krishna	Design the backup module and implement file copying using open (), read (), write (), and close ().
2520090099	Chaithanya	Implement directory traversal, file comparison, and synchronization logic using stat (), opendir (), and readdir ().
2520090218	Navaneeth	Handle testing, error handling, documentation, GitHub repository, and project demonstration.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment for Linux system calls.
C	Main programming language for implementing the backup and synchronization system.
GCC	Compiles the C source code into an executable program.
VS Code / Vim	Used to write and edit the C source files.
Git / GitHub	Version control and storage of the project source code.

8. Expected Outcome

The project is expected to produce a working command-line Linux application that can back up files from a source directory and synchronize new or modified files with a backup directory. It will use Linux/POSIX system calls for file access and demonstrate file descriptors, directory handling, metadata comparison, permissions, and error handling. The program will display the files processed and report successful copies or errors. The final system will provide a practical demonstration of how Operating Systems system calls can be used to implement a real-world file backup and synchronization task.

9. GitHub Repository

GitHub Repository Link: https://github.com/chaitanyakonninti07-max/klh_CSIT_2026_22_Linux-file-backup-and-synchronisation-system-using-system-calls

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Ragupathi.M

Signature:

Remarks:

Faculty Use

Project Approved:

☐ Yes ☐ No

Date:

Faculty Signature: